Squaring extends to a unique function on the Cauchy reals with the expected Lipschitz bound

The Univalent Foundations Program

hott-book:reals.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'RC-squaring' (source lines 1636-1646, Theorem 11.3.46); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 11, section 3, theorem offset 45) make the counted theorem print as Theorem 11.3.46, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The five external references are navigation pointers, all with dump-confirmed numbers; every one of them is invoked by label inside the retained proof, and none replaces any part of the counted proof, which is retained in full.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: extension of squaring to the Cauchy reals with its Lipschitz bound (Theorem 11.3.46).

Allowed standard prerequisites (editorial).

- Archimedean property of the Cauchy reals. Every real is bounded in absolute value by some natural number.. It makes the map from Sigma n, [-n,n] to RC surjective at source lines 1649-1658.
- Extension of Lipschitz maps. A Lipschitz map on the rationals extends to a Lipschitz map on the reals with the same constant.. It is applied to the squaring map on each bounded interval at source lines 1666-1668.
- Uniqueness of factorisation through a surjection. A map respecting the kernel pair of a surjection factors uniquely through it.. It produces the unique function at source lines 1671-1679.

Source reference. Original exercise 11.7: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1411–1411. Original result cited at a distance in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot; a pointer is used rather than a context span because the cited result lies in another section or is separated by unre-tained environments, so a retained context would print with the wrong number.

Source reference. Original lemma 11.3.15: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1578–1578. Original result cited at a distance in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot; a pointer is used rather than a context span because the cited result lies in another section or is separated by unre-tained environments, so a retained context would print with the wrong number.

Source reference. Original exercise 11.8: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1582–1582. Original result cited at a distance in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot; a pointer is used rather than a context span because the cited result lies in another section or is separated by unre-tained environments, so a retained context would print with the wrong number.

Source reference. Original lemma 11.3.39: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1577–1577. Original result cited at a distance in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot; a pointer is used rather than a context span because the cited result lies in another section or is separated by unre-tained environments, so a retained context would print with the wrong number.

Source reference. Original lemma 10.1.5: sources/hott-reals/book-labelnumbers-first-edition.tex, source lines 1541–1541. Original result cited at a distance in the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot; a pointer is used rather than a context span because the cited result lies in another section or is separated by unre-tained environments, so a retained context would print with the wrong number.

Theorem 11.3.46. *There exists a unique function* $(-)^2 : \mathbb{R}_\mathsf{c} \to \mathbb{R}_\mathsf{c}$ *which extends squaring* $q \mapsto q^2$ *of rational numbers and satisfies*

$$\forall(n : \mathbb{N}).\, \forall(u, v : [-n, n]).\, |u^2 - v^2| \leq 2 \cdot n \cdot |u - v|.$$

Proof. We first observe that for every $u : \mathbb{R}_\mathsf{c}$ there merely exists $n : \mathbb{N}$ such that $-n \leq u \leq n$, see original exercise 11.7, so the map

$$e : \Big(\sum_{n:\mathbb{N}} [-n, n]\Big) \to \mathbb{R}_\mathsf{c} \qquad \text{defined by} \qquad e(n, x) :\equiv x$$

is surjective. Next, for each $n : \mathbb{N}$, the squaring map

$$s_n : \{\, q : \mathbb{Q} \mid -n \leq q \leq n \,\} \to \mathbb{Q} \qquad \text{defined by} \qquad s_n(q) :\equiv q^2$$

is Lipschitz with constant $2n$, so we can use original lemma 11.3.15 to extend it to a map $\bar{s}_n : [-n, n] \to \mathbb{R}_\mathsf{c}$ with Lipschitz constant $2n$, see original exercise 11.8 for details. The maps $\bar{s}_n$ are compatible: if $m < n$ for some $m, n : \mathbb{N}$

then s_n restricted to $[-m, m]$ must agree with s_m because both are Lipschitz, and therefore continuous in the sense of original lemma 11.3.39. Therefore, by original lemma 10.1.5 the map

$$\left(\sum_{n:\mathbb{N}} [-n, n]\right) \to \mathbb{R}_{\mathsf{c}}, \qquad \text{given by} \qquad (n, x) \mapsto s_n(x)$$

factors uniquely through $\mathbb{R}_{\mathsf{c}}$ to give us the desired function. □